



Seedling drought tolerance and functional traits vary in response to the timing of water availability in a keystone Hawaiian tree species

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Abstract Climate change models project an increase in the frequency and duration of drought globally. Changes in rainfall are expected to have particularly detrimental effects on seedlings due to their inability to reach deep water sources. We conducted a greenhouse experiment to test how the timing of watering affects seedling performance, physiology, and morphology in a keystone island endemic tree, *Metrosideros polymorpha*. Seedlings were grown for 60 days in one of five conditions: control treatment with daily watering to 100% field capacity (FC), a 28-day pulse drought with no water, and three groups maintained at 80% FC and watered every 2, 7, or 14 days. Seedling drought tolerance depended on the duration of drought; growth for plants watered to 80% FC every 2 or 14 days was comparable to 100% FC plants, but was reduced in the 28-day pulse drought

and 7-day 80% FC. Survival was high in all treatments except the pulse drought. Stomatal conductance, chlorophyll content, and C:N ratio were positively related to height and biomass under drought. Traits associated with resource investment, fluorescence, and water use were plastic in their expression, shifting under drought compared to control conditions. *M. polymorpha* seedlings exhibit trait plasticity that allows them to tolerate drought, but its duration and intensity influence their survival and growth. Our results highlight the need for experimental studies manipulating temporal variation in water with explicit consideration of the role of trait plasticity in drought tolerance.

Keywords Climate change · Endemic species · Island · *Metrosideros polymorpha* · Trait plasticity

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Introduction

Soil-moisture availability has been shown to strongly influence the growth and mortality of tree seedlings in forest ecosystems around the world (Fisher et al. 1991; Veenendaal et al. 1996; Engelbrecht et al. 2006; Poorter and Markesteijn, 2008). Even in tropical forests, where rainfall is high (800 to 10,000 mm), temporal variation in rainfall associated with seasonality can impose significant drought stress (Corlett 2016). Tree seedlings are more vulnerable to drought than other life stages (Gerhardt 1996; Marod et al. 2002) because their under-developed root systems limit access to deeper and more stable sources of soil water during periods of low moisture. Simultaneously, climate change is altering the frequency and intensity of rainfall, even in aseasonal equatorial rainforests (Feng et al. 2013; Chadwick et al. 2016). The need to understand and predict how climate change will influence forest communities has resulted in a significant increase in the number of studies assessing drought tolerance for tropical tree seedlings (Engelbrecht and Kursar 2003; Poorter and Markesteijn 2008; O'Brien et al. 2013, 2015). Many such studies manipulate the magnitude but not the temporal variation in water (Bonafant et al. 2016), even though the timing of rainfall events can significantly influence seedling recruitment (O'Brien et al. 2015).

Climate change models project shifts in the frequency and duration of drought across tropical forest ecosystems. In Hawaii, climate models project more days without rain, less precipitation overall (Timm et al. 2015; Frazier and Giambelluca 2016), more extreme precipitation events (i.e., storms) on the islands of Oahu and Kauai (Chu et al. 2010; Chu and Chen 2005; Chen and Chu 2014; Zhang et al. 2016), and increased temperatures in native forests at high elevations (Giambelluca et al. 2008). Changes in sea surface temperature will likely increase the frequency and intensity of hurricanes reaching Hawaii (Wendland 1977; Emanuel 1987; Murakami et al. 2013), altering the canopy structure of Hawaii's forests and shifting the demographic composition to younger trees. Insights into drought-mediated shifts in forest composition and structure are thus essential for maintaining future ecosystem stability in Hawaii.

The dominant native canopy tree species across Hawaii's forests is *Metrosideros polymorpha* (Myrtaceae), known in Hawaiian as 'ōhi'a lehua, a slow-

growing, endemic evergreen tree. Adult *M. polymorpha* provide habitat for endemic invertebrates (Gagne 1979), nectar for endemic birds (Pimm and Pimm 1982), and function as hosts for epiphytes, mosses, and seedlings of subcanopy shrubs and other *M. polymorpha* (Santiago 2006). In addition to its ecological significance, *M. polymorpha* is also culturally significant. Many Hawaiian legends mention 'ōhi'a lehua and its forests as sacred to the volcano goddess Pele. *M. polymorpha* occurs across a broad ecological gradient (from mountain ridges to coastlines) and a large precipitation gradient (from ca. 400 to 10,000 mm rain per year) (Price et al. 2012), and exhibits polymorphisms within and among forest populations (Stemmermann 1983; Joel et al. 1994; Kitayama et al. 1997; DeBoer and Stacy 2013). The stand dynamics of *M. polymorpha* have been well studied since the 1970s, yet much remains unknown about how it regenerates (Mueller-Dombois 1983; Burton and Mueller-Dombois 1984; Jacobi et al. 1988; Drake 1993; Boehmer et al. 2013), including which traits are under strongest selection at the seedling stage (but see Morrison and Stacy 2014). The need to understand recruitment in *M. polymorpha* is greater than ever before, due to ongoing large-scale mortality events among adults, resulting from fungal pathogens that cause "rapid 'ōhi'a death" (Keith et al. 2015; Mortenson et al. 2016). Because the *Metrosideros* genus is widely distributed across Oceania, occurring in tropical and temperate regions (Wilson 1996), our results are relevant to other forests throughout the south Pacific.

Predicting climate change effects on the stand dynamics of *M. polymorpha* will benefit from knowledge of how temporal variability in rainfall will influence growth, survival, and plant traits in seedlings. We focus on traits associated with water relations, carbohydrate production, and resource allocation strategy, as these often determine a plant's ability to withstand drought. For example, under periods of drought, leaf traits such as specific leaf area (SLA) and transpiration may decrease, while root:shoot biomass and instantaneous water-use efficiency (WUE) generally increase in drought-tolerant species (Hamann et al. 2018). In mature *M. polymorpha* trees, specific leaf area, leaf size, hydraulic conductivity, and pubescence have been shown to vary strongly in response to moisture availability (Geeske et al. 1994; Cordell et al. 1998; Cornwell et al. 2007), which may

explain *M. polymorpha*'s dominance across the landscape. Because morphological and ecophysiological traits vary with ontogeny (Mediavilla and Escudero 2004; Barton and Boege 2017; Damián et al. 2018), and the influence of traits on plant performance also changes across life stages (Gibert et al. 2016), it is difficult to extrapolate results from studies conducted on adults to juvenile stages. Indeed, more work is needed to understand how leaf traits influence seedling growth and survival.

Our aim was to test how drought severity and the frequency of watering affect *M. polymorpha* seedling performance and traits, using a greenhouse experiment. We sought to answer three questions: (1) How tolerant are seedlings of *M. polymorpha* to reductions in total water? We include three types of sustained drought where total water was reduced relative to control, and a 28-day drought where no water was provided at all, hereafter press and pulse droughts, respectively. Seedlings that are drought sensitive may experience reduced growth, reduced survival, or both, according to Larcher (2003) and Engelbrecht and Kursar (2003). (2) How are *M. polymorpha* seedling's survival and growth (i.e., performance) influenced by the duration of drought? Here we compared the press-drought groups to one another. (3) How do morphological and physiological traits change in response to the watering treatments, and how does this influence seedling performance? For example, WUE might positively correlate with plant performance under drought, but not under well-watered conditions. Such a pattern would indicate that WUE is an important mechanism of drought tolerance in *M. polymorpha* seedlings.

We predicted that *M. polymorpha* seedling survival and growth would be quite tolerant to drought, as the individuals used in this study originated from a dry forest population from the lower limit of *M. polymorpha*'s precipitation range (see "Methods"). Given the relatively dry habitat from which our seeds were collected and considering recent evidence for local adaptation in seedling drought tolerance in this species (Barton et al. in review), the focal population is likely to exhibit drought adaptation. Moreover, we predicted that *M. polymorpha* seedlings would express phenotypic plasticity in traits conferring drought tolerance. The results of this study will provide a more mechanistic understanding of how physiological and morphological traits are interlinked for seedlings exposed

to drought and may inform models regarding future range shifts for keystone tree species. We expand upon previous studies assessing the influence of variation in watering on seedling traits and performance (O'Brien et al. 2013, 2014, 2015) by measuring a larger number of morphological and physiological traits than have previously been included. In light of climate change, differences in species' tolerances to drought at the seedling stage will be a major determinant of future species distributions and ecosystem productivity in tropical forests (Fortunel et al. 2014), highlighting the need for studies that describe the underlying traits conferring or constraining drought tolerance.

Methods

Experimental setup and conditions

Seeds were collected from a high-elevation dry forest population near Mauna Kea on Hawaii Island in December 2015. Mean annual precipitation at the site of collection is ca. 500 mm. Dry forest populations are of particular interest because while they are likely to exhibit high drought tolerance, they occur in habitats most at risk of future increases in the frequency and intensity of drought in Hawaii (Timm et al. 2015). There are five recognized varieties on Hawaii Island, and evidence of genetic differentiation among four of these five (Stacy et al. 2014). However, given the small spatial scale at which the seeds were collected, only a single variety is likely to be represented in the sample population.

Metrosideros polymorpha seeds are minute, and multiple fruits (e.g., thousands of seeds) were collected from at least 10 mature trees. Seeds were mixed together to produce a genetically mixed sample population. Seeds were sown in flat containers filled with a 1:1 ratio by volume of Pro-Mix BX (Pro-Mix[®], Premier Tech., Rivière-du-Loup, Quebec, Canada) to black cinder in February 2016 at the University of Hawaii at Manoa (Honolulu, Hawaii, USA) Pope Greenhouse facility. After germination, seedlings were transplanted at the four-leaf stage into 3.78-L pots filled with a 1:1 volume of Pro-Mix BX:black cinder and treated with a single application of slow-release fertilizer (Osmocote Plus 15-9-12, Scotts Co., Marysville, Ohio, USA) at the time of transplant. Transplant day marks day 0 (D0) of the experiment.

All seedlings were watered daily for 13 days. On day 14, plants were randomly separated into five treatment groups that differed in the timing and amount of watering: the “control” group was watered daily until saturation and assumed to be at 100% field capacity (“100FC”); three different groups that experienced a press drought such that water was applied to achieve 80% field capacity (using gravimetric water content) (“80FC”) based on the control pots (Sack 2004), and these groups differed in watering frequency: one group received water every two days (“2D”), another group received water every 7 days (“7D”), and the third group received water every 14 days (“14D”). Thus, plants under the 80FC treatments all received the same total amount of water over the study period, but seedlings that were watered more frequently received less water during each watering event than those that were watered less frequently. The fifth watering treatment was a 28-day pulse drought during which time no water was applied, and after which, plants were watered daily to 100% FC. We used gravimetric water-content assessment to maintain plants in the 80FC groups at 80% field capacity throughout the experiment (D14–D69 when harvested). At each watering event, a subset of 20 pots were weighed to calculate the % field capacity, and the amount of water required to bring them back up to 80% FC was calculated and added to each pot by hand. Thus, the amount of water added to the 80FC pots increased over the course of the experiment as plants used more water, and the pots dried out more between watering events. This press-drought level was chosen to impose sufficient drought stress while maintaining low mortality, based on a previous study (K. Barton, University of Hawaii at Manoa, unpublished data). There were 20 replicate plants per watering treatment at the onset of the experiment, and all seedlings had reached the 6–8 leaf stage at the time the watering treatments were initiated. In addition to the gravimetric water-content calculations during watering events, volumetric soil water content (%) was measured using a HydroSense II soil-moisture probe (Campbell Scientific, Logan, UT, USA) on all plants every week starting on D7. We initiated the collection of all trait data on D14. The experiment concluded on D69 when final traits were measured, and plants were harvested (Appendix 1). To minimize positional effects within the greenhouse, the pots were re-randomized within benches weekly.

Seedling performance

A total of 18 response variables were measured for the plants in each group (Appendix 2 lists variables and sample sizes, and Appendix 3 lists details of response variables), which fall into three categories: performance metrics, leaf physiological traits, and morphological traits. Performance was assessed with survival, height, and total biomass. Survival was determined daily based on greenness of aboveground tissue as well as brittleness of leaves and stems. Height (cm) was measured on all plants once per week starting on D35. Total dry plant biomass (g) was measured for plants harvested at the end of the experiment. We separated above- and below-ground tissues, and dried them for 7–14 days at 60 °C. We weighed shoot and root tissues, which were summed to compute total biomass.

Leaf physiological traits

Leaf stomatal conductance ($\text{mmol m}^{-2} \text{s}^{-1}$) was measured on all plants weekly starting on D14, up to seven times over the course of the experiment using an SC-1 steady state leaf porometer (Decagon Devices, Pullman, Washington, USA). All other physiological traits were measured once on D69 before plants were harvested, including: net photosynthesis ($\mu\text{mol CO}_2 \text{m}^{-2} \text{s}^{-1}$), transpiration ($\text{mmol H}_2\text{O m}^{-2} \text{s}^{-1}$), and instantaneous water-use efficiency (net photosynthesis/transpiration) using a LICOR 6400 XT (LI-COR, Lincoln, Nebraska, USA) on the youngest fully mature leaf (the leaf was randomly selected if more than one was available). Across all physiological measurements, photosynthetically active radiation was set to $1500 \mu\text{mol photons m}^{-2} \text{s}^{-1}$, ambient CO_2 concentration was set to $400 \mu\text{mol mol}^{-1}$, mean leaf temperature was 37 °C (35–38), and mean relative humidity was 45% (44–48). For a subset of individuals, we also measured various fluorescence parameters using a MINI-PAM II fluorometer (Walz, Munich, Germany). These parameters describe the efficiency with which plants process absorbed irradiance, and are useful indicators of light and drought stress: quantum efficiency of the photosystems (α , electrons/photons), maximum non-photochemical quenching (NPQ_{max} , relative units), maximum electron transport capacity (ETR_{max} , $\mu\text{mol electrons m}^{-2} \text{s}^{-1}$), minimum

saturation irradiance (E_k , $\mu\text{mol photons m}^{-2} \text{ s}^{-1}$), and quantum yield (Y_0 , $\text{mol CO}_2 \text{ mol photons}^{-1}$). We used a SPAD 502 Plus Chlorophyll Meter (Spectrum Technologies, Inc, Aurora, Illinois, USA) to measure leaf chlorophyll content (indicated by degree of greenness, relative units) in two leaves per individual and used the mean as our response variable. All physiological traits are area-based. Due to the high mortality and exceptionally small size of plants in the pulse-drought treatment, we were unable to quantify photosynthetic rates (leaves were too small for the LICOR6400 XT but large enough for the MINI-PAM II) and were only able to estimate fluorescence parameters for a subset of the seedlings (see Appendix 4 for photographs of seedlings).

We measured long-term WUE using carbon stable isotopes ($\delta^{13}\text{C}$ values), where less negative $\delta^{13}\text{C}$ values indicate lower time-integrated c_i/c_a (intercellular $[\text{CO}_2]$: atmospheric $[\text{CO}_2]$) and higher water-use efficiency. Leaves that continue to photosynthesize while stomata are closed (high instantaneous WUE) exhibit greater drawdown of intercellular CO_2 concentration (c_i) relative to atmospheric CO_2 (c_a), and less discrimination against heavy carbon isotopes. Tissues were dried for 7–14 days at 60 °C. We removed all leaf tissue from ten oven-dried replicate seedlings from each treatment group, ground each replicate separately in a MM200 ball mill (Retsch, Inc., Newton, Pennsylvania, USA) at 25 Hz for 60–90 s and passed all material through a 250- μm sieve. Samples (0.8–0.9 mg) were packed into tin capsules and analyzed on a Costech ECS 4010 Elemental Analyzer coupled to a Picarro G2201-i cavity ringdown spectrometer (Picarro, Santa Clara, California, USA), which provided leaf nitrogen (%N), C:N ratios (unitless), and $\delta^{13}\text{C}$ (in units of permil, ‰, vs. VPDB standard) of the leaf tissues. Percent leaf nitrogen content was strongly and negatively correlated with C:N ($r = -0.98$, $P < 0.0001$).

Morphological traits

We used a micrometer to measure leaf thickness (mm) at the end of the experiment on all living plants. The same leaf, which was used to estimate net photosynthesis, transpiration, and instantaneous water-use efficiency, was collected and photographed, and we used ImageJ v. 1.51j (U.S. National Institutes of

Health, Bethesda, MD, USA; Rasband 1997–2016) to compute leaf area (cm^2). It was then dried in an oven for 7–14 days at 60 °C, and weighed (g) using an analytic balance. Leaf area was divided by the oven-dried mass to compute specific leaf area, SLA ($\text{cm}^2 \text{ g}^{-1}$).

Statistical analyses

Kolmogorov Smirnov tests were used to test whether the temporal patterns of soil water content varied significantly among treatment groups. This analysis tests the null hypothesis that two vectors (soil water content readings from two different treatments) are drawn from the same continuous distribution. A one-way ANOVA was used to determine if the mean soil water content measurements of the treatment groups differed, and Tukey HSD was used to analyze pairwise differences across groups (Appendix 5).

To test how the watering treatments influenced the performance of *M. polymorpha* seedlings, we conducted a one-way ANOVA on total biomass and height, separately. Total biomass and height were log-transformed to achieve normality. A Tukey HSD test was performed to determine which pairwise comparisons were statistically significant. We used ANOVA to test whether seedling height during the early stages of the experiment (day 35) varied by the treatment group and found no significant differences (Appendix 6). We also calculated the change in height as height at day 69 minus height on day 35 for a subset of all individuals ($N = 37$), to determine whether relative growth rates differed among groups (Appendix 6). Mean height (on day 35) per treatment group was used as a covariate when testing for differences in size across groups, but our results were unchanged, and therefore we show only the results of analyses that do not include the covariate. A Chi-square goodness-of-fit was used to determine whether the proportion of deaths (i.e., survival) among the watering treatments differed more than expected due to chance alone. We also used a failure time analysis to determine the median number of days until death for the pulse-drought treatment but not the remaining treatments due to low mortality (not shown).

To test for plasticity in the physiological and morphological traits in response to the watering treatments, we conducted a series of univariate

analyses with the main predictor variable of treatment. We examined the traits individually, rather than jointly, because we were interested in how each trait responded to the treatments, and because MANOVA requires moderate correlations among all traits (French et al. 2008), which was not true for all pairwise comparisons among the physiological ($0.03 < r < 0.51$) or morphological ($0.08 < r < 0.22$) traits (Appendix 7). Traits were transformed (via log or square root) as needed, to meet statistical assumptions of normality. α (the efficiency of the photosystems) could not be transformed to achieve normality, and therefore we tested for the effects of watering treatments using a Kruskal–Wallis test. We used Tukey HSD tests to analyze pairwise differences across groups when the effect of drought was significant.

Finally, to test how the physiological traits influenced seedling growth performance (total biomass and height) under different watering treatments, we used ANCOVA with watering treatment as the predictor variable and trait as the covariate. We conducted a separate ANCOVA for each physiological trait. For models where we detected significant interactions, we discuss performance among groups only for individual regression models that are significant (Appendix 8). The low predictive power of our statistical models likely reflect the low sample sizes in this experiment, although we still detected significant relationships for several traits. There were insufficient data to test how chlorophyll content and NPQ_{max} influenced growth performance in plants under pulse drought (see “Results”).

We did not test for interactions among more than two physiological variables at a time, due to the difficulty in interpreting these results, and because our sample sizes were not large enough to accurately assess all possible n -way interactions among the variables. For stomatal conductance and height, which were measured multiple times throughout the experiment, we used linear mixed effects models with treatment as the fixed effect and individual plant nested within date as random effects. We found that the effect of treatment was statistically significant when both methods were used (height, $F_4 = 14.3$, $P < 0.0001$; stomatal conductance $F_4 = 26.6$, $P < 0.0001$). Therefore, we only present the results from the ANOVA and ANCOVA for these variables. All statistical analyses were done using R version 3.4.3 (R Core Team 2013). Figures 2, 3, 4, and 5 were

produced using ggplot (`‘ggplot2’` package; Wickham 2009). Linear mixed models were analyzed using the lme function (`‘nlme’` package). The failure time analysis was conducted with the survfit function, using the Kaplan–Meier method.

Results

Treatment effects on water availability

The watering treatments significantly influenced the temporal patterns of soil water content (Fig. 1, Kolmogorov–Smirnov, all $P < 0.001$). As expected, soil water content was similar for all plants on day 13 of the experiment, before the watering treatments were imposed. Reduction in soil water under the press drought (80FC) and pulse-drought (PD) treatments quickly led to a decrease in soil water content with the PD pots being the driest by day 30. There was an unexpected peak in soil water content on day 51 of the experiment, which occurred across all treatments, and this resulted from a reduction in ambient temperature and increased humidity within the greenhouse (data

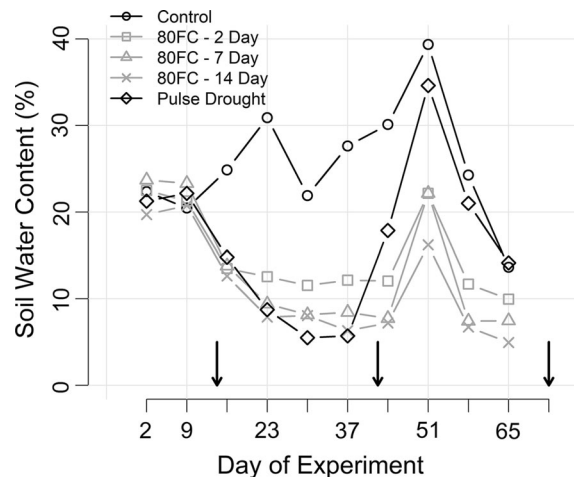
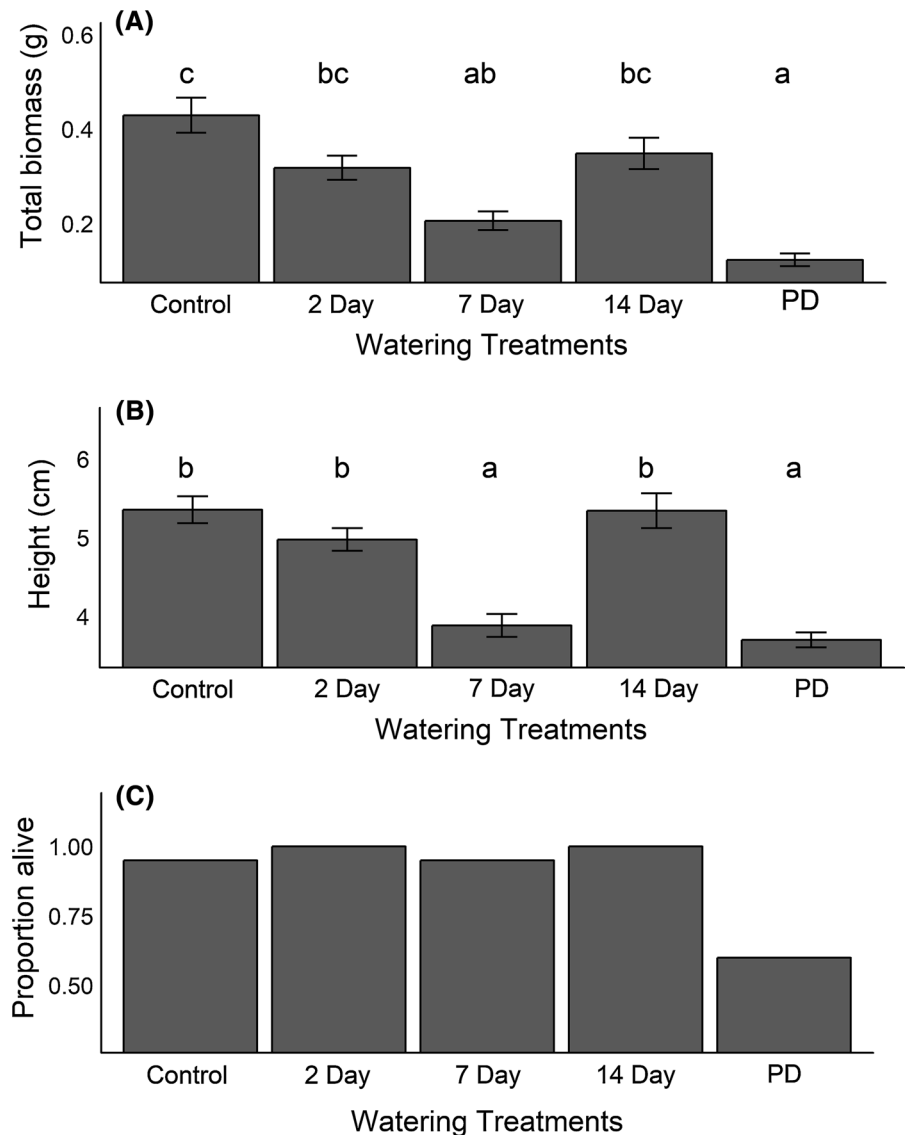


Fig. 1 Temporal trends in soil water content among control and treatment groups for *M. polymorpha* seedlings, as estimated during the experiment. The control group (black line, circle) was watered daily to 100% field capacity (100FC), while three groups were watered to 80FC (gray lines) but differed in the timing of watering (2 day, 7 day, and 14 days—square, triangle, and cross, respectively), and one group experienced 28-days with no water, i.e., pulse drought (black line, diamond). Arrows indicate significant time points during experiment: first arrow is the separation into treatment groups, second arrow is the end of pulse drought, and final arrow is the harvest (end of experiment)

Fig. 2 Effect of watering treatments on **a** total biomass, **b** height, and **c** survival of *M. polymorpha* seedlings. The control group was watered daily to 100% field capacity (100FC); the 2 day, 7 day, and 14 day groups were watered to 80FC; the pulse drought group ('PD') received no water for 28 days. Lowercase letters indicate statistically significant differences across groups, using Tukey HSD. Means $\pm$ SE are shown for all traits except survival. We did not compute pairwise differences for survival



not shown). Soil water content in the pulse drought pots approached that of the control pots prior to day 51, as both were watered daily between days 42–65. Watering frequency significantly influenced mean soil water content for the three groups maintained at 80FC (Tukey's $P_{\text{all comparisons}} < 0.05$, Appendix 5). This suggests that soil water content of the pots that were watered less frequently (7D or 14D pots) could not recover to levels comparable to pots watered more frequently (2D pots), possibly due to variation in rewetting potential (O'Brien et al. 2013, 2015).

Influence of watering on seedling performance

Seedling performance (survival, total biomass, and height) differed significantly due to the watering treatments. Survival of seedlings throughout the experiment differed significantly across treatments ($\chi^2 = 12.778$, $df = 4$, $P = 0.01$, Fig. 2c), and was nearly halved in the PD group compared to the control (100FC) group. Survival was high among all press drought (80FC: 2-, 7-, and 14-day)-treatment groups and the control plants, as we only observed two deaths (one in the control and one in the 7-day group). For

plants in the PD group, the median time until death was 34 days.

For surviving seedlings, total dry weight biomass differed significantly among the watering treatments ($F_{4,84} = 4.8$, $P = 0.002$, Fig. 2a) and was 3.5 times higher in the control group compared to the PD group. Total biomass for press-drought groups was similar to that of control plants when seedlings were watered every 2 or 14 days (Fig. 2a). Seedling height also differed significantly due to the watering treatments ($F_{4,295} = 15.45$, $P < 0.0001$, Fig. 1b), and the pattern mirrored that of total biomass. Mean seedling height was 40% lower in the PD group compared to the control group, but height differences within the press drought groups were due to the initial variations in size (Appendix 6, Fig. 2b). Relative growth rates in terms of height were no different among the 80FC groups ($F_{4,33} = 1.74$, $P = 0.17$), and these were all significantly lower than the control group (Tukey $P < 0.05$). Together, these performance metrics reveal low tolerance to a sustained 28-day pulse drought in *M. polymorpha* seedlings and high tolerance to the 80% press drought, with minimal variation due to timing of watering.

Influence of watering on physiology and morphology

The watering treatments significantly influenced 8 of the 12 physiological traits and 2 of the 3 morphological traits (see Appendix 9, for traits that were not influenced by the watering treatments), revealing plasticity in various traits related to drought tolerance and vigor. Specific rankings among drought groups varied (Fig. 3), but we observed that compared to control seedlings, drought reduced stomatal conductance in both press and pulse droughts groups (Tukey $P < 0.05$), and, for pulse drought plants, increased instantaneous water-use efficiency and leaf chlorophyll (Tukey $P < 0.05$). Significant variation among treatment groups were observed for stomatal conductance ($F_{4,311} = 26.51$, $P < 0.001$), water-use efficiency ($F_{3,10} = 8.09$, $P = 0.005$), % N ($F_{4,44} = 2.97$, $P = 0.03$), $\text{NPQ}_{\max}$ ($F_{4,64} = 3.27$, $P = 0.02$), $\text{ETR}_{\max}$ ($F_{4,64} = 2.93$, $P = 0.03$) and E_k ($F_{4,64} = 2.85$, $P = 0.03$). Overall, with increasing drought severity (progressing from control to pulse drought) stomatal conductance decreased while WUE increased. The

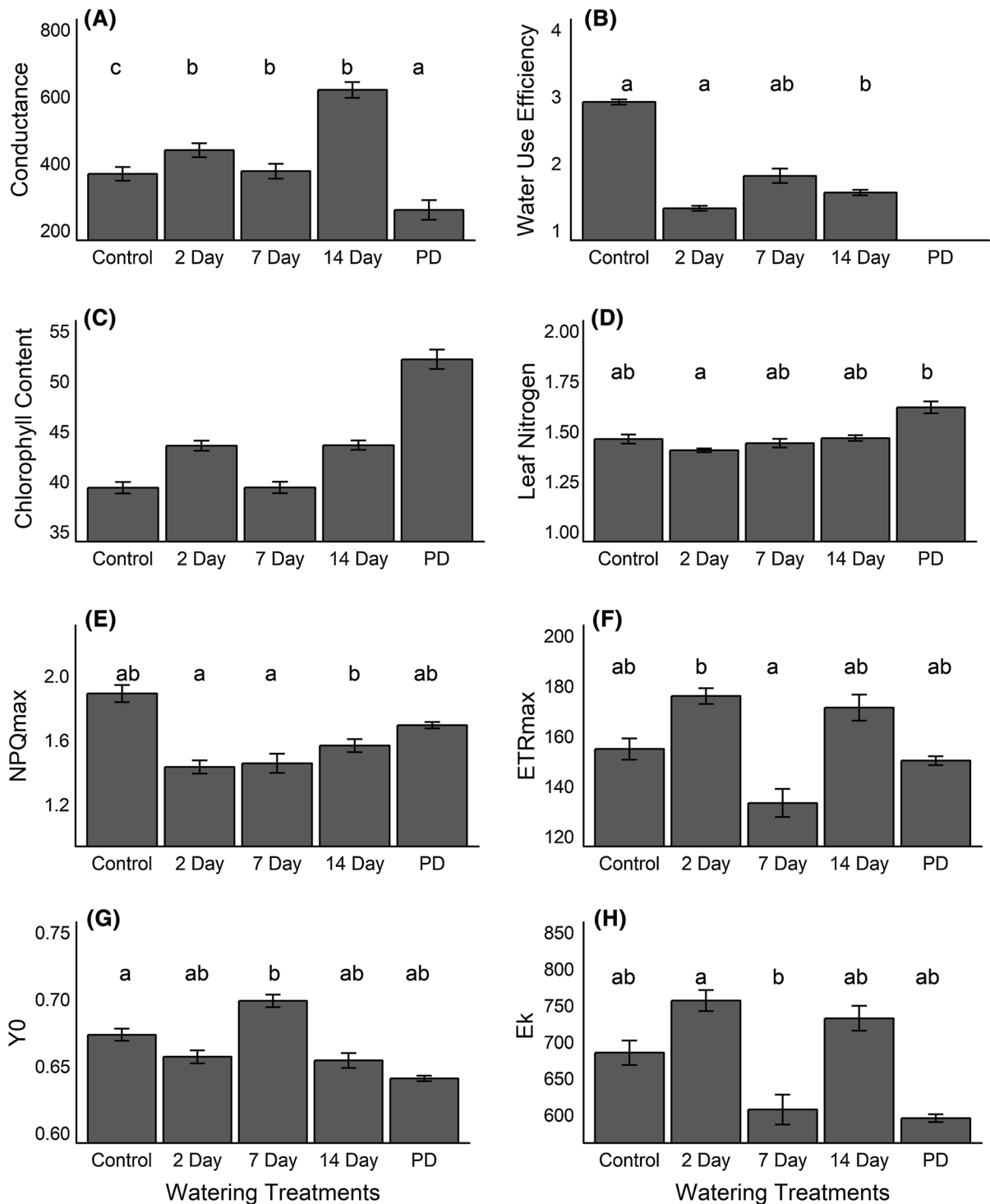
Fig. 3 Effect of watering treatments on physiological traits A–H for *M. polymorpha* seedlings. The control group was watered daily to 100% field capacity (100FC); the 2 day, 7 day, and 14 day groups were watered to 80FC; the pulse drought ('PD') group received no water for 28 days. There was a main effect on watering on chlorophyll content but the pairwise differences were not significant. Means $\pm$ SE are shown for all traits. Lowercase letters indicate statistically significant differences across groups, using Tukey HSD. Units: Stomatal conductance, $\text{mol H}_2\text{O m}^{-2} \text{ s}^{-1}$; water-use efficiency, $\mu\text{mol CO}_2 \text{ mmol}^{-1} \text{ H}_2\text{O}$; chlorophyll content, unitless; leaf nitrogen, %; $\text{NPQ}_{\max}$ ($F_m^0 - F_m'/F_m'$); $\text{ETR}_{\max}$, $\mu\text{mol electrons m}^{-2} \text{ s}^{-1}$; Y_0 , $\text{mol CO}_2 \text{ mol photons}^{-1}$; E_k , $\mu\text{mol photons m}^{-2} \text{ s}^{-1}$

following traits did not respond to the watering treatments: photosynthesis ($F_{3,10} = 1.15$, $P = 0.38$), transpiration ($F_{3,10} = 2.21$, $P = 0.15$), α (indicator of photosynthetic efficiency, $\chi^2(4) = 3.25$, $P = 0.52$), and $\delta^{13}\text{C}$ (indicator of long-term water-use efficiency, $F_{4,44} = 1.09$, $P = 0.37$).

Plasticity in response to drought was also detected for morphological traits. Drought was associated with reduced leaf thickness in the pulse-drought seedlings ($F_{4,64} = 2.6$, $P = 0.04$, Fig. 4), although the pairwise comparisons were not statistically significant (Tukey $P > 0.05$). Biomass allocation responded to drought with a significant shift toward root biomass for seedlings in the pulse drought and 14-day press drought ($F_{4,83} = 5.38$, $P = 0.0007$, Fig. 4). Drought had no effect on specific leaf area ($F_{4,84} = 0.66$, $P = 0.62$, Appendix 9). Root biomass increased with increasing drought when we controlled for differences in seedling size ($F_{4,83} = 5.17$, $P = 0.0009$, Fig. 4).

Physiological mechanisms underlying drought tolerance

We found evidence of significant interactions between some leaf physiological traits and the watering treatments on seedling growth, revealing these to be traits that could influence fitness under drought conditions in *M. polymorpha* (Fig. 5). There was a significant interaction between stomatal conductance and watering treatment on seedling height ($F_{4,290} = 3.26$, $P = 0.01$). Seedlings with a greater mean stomatal conductance grew taller on average than those with lower conductance values; the magnitude of the effect was greatest for the 14-day group and lowest for plants watered every two days (Fig. 5, Appendix 8). We



found a marginally significant interaction between chlorophyll content and watering treatment on

seedling height ($F_{3,51} = 2.69$, $P = 0.06$). High chlorophyll content was associated with taller plants in the 7-

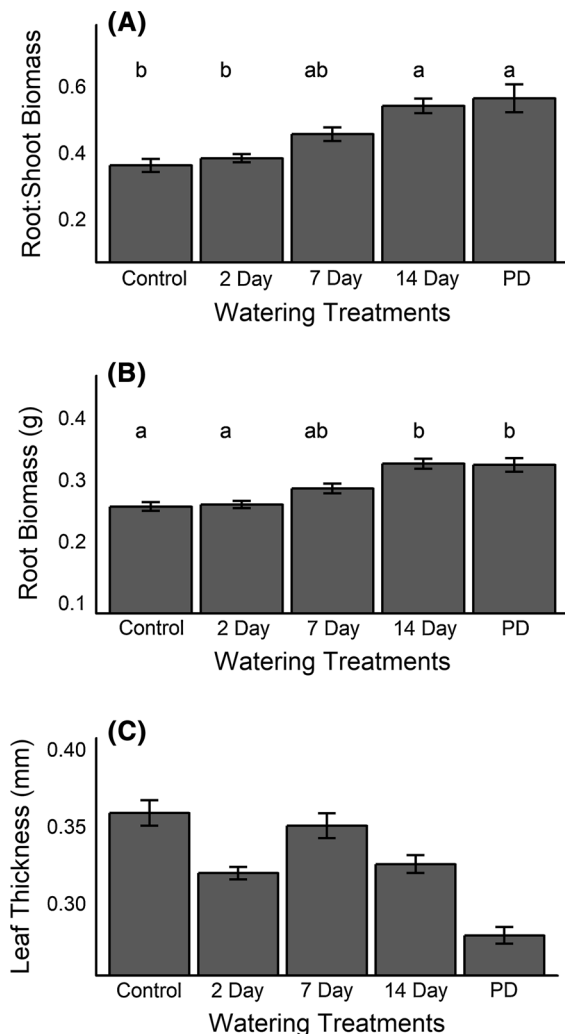


Fig. 4 Effect of watering treatments on morphological traits A–C for *M. polymorpha* seedlings. The control group was watered daily to 100FC; the 2 day, 7 day, and 14 day groups were watered to 80FC; the pulse drought (‘PD’) group received no water for 28 days. Despite a significant effect of watering treatment on leaf thickness (B), we did not detect any pairwise differences, therefore, letters are not shown. Means $\pm$ SE are shown for all traits

and 14-day groups but the magnitude of the effect was greater in the 7-day group (Appendix 8). These findings indicate that high chlorophyll investment promotes drought tolerance in *M. polymorpha* under press drought conditions (Fig. 5).

In addition, there was a marginally significant interaction effect between C:N and watering treatment on seedling total biomass ($F_{4,39} = 2.28$, $P = 0.08$), but no interaction between percent leaf nitrogen and

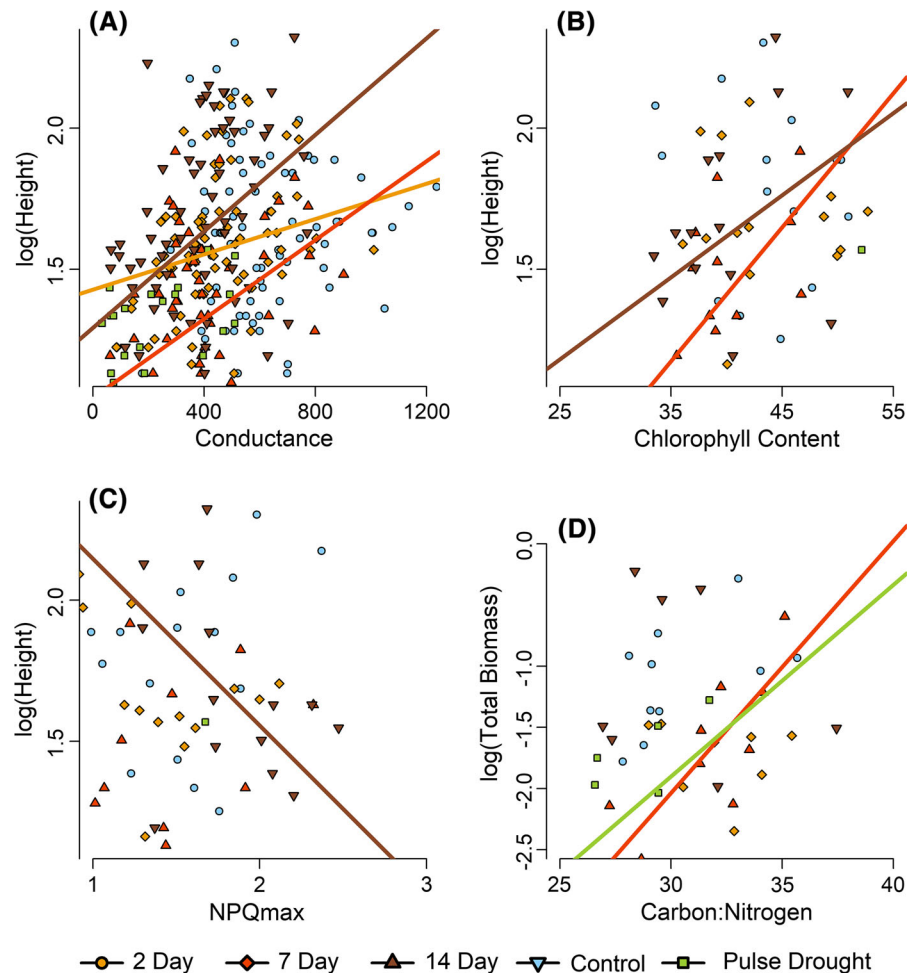
biomass ($F_{4,40} = 1.74$, $P = 0.16$) or height ($F_{3,25} = 0.89$, $P = 0.46$). A high C:N was found among seedlings with a high total biomass when they were watered every 7 days (Fig. 5, Appendix 8), and to a lesser extent, we observed the same pattern in the pulse drought group. C:N appeared to slightly constrain growth performance under a pulse drought when compared to plants watered every 7 days. While leaf nitrogen per se did not contribute to differential performance among the treatment groups, the amount of nitrogen relative to carbon does appear to play a role in mediating performance under drought for *M. polymorpha* seedlings. Resource allocation strategies, such as shifts in investment from nitrogen-poor to nitrogen-rich leaves, thus warrant further investigation.

Finally, we found a significant interaction effect between NPQ_{max} and watering treatment on seedling height ($F_{3,52} = 3.44$, $P = 0.02$). In contrast to the previous interactions, high NPQ_{max} was associated with reduced height but only in seedlings watered every 14 days, whereas in the remaining treatments, seedling height did not vary with NPQ_{max} (Appendix 8, Fig. 5). These findings indicate that NPQ_{max} imposes constraints on the seedling growth performance but only under a moderately severe drought.

Discussion

This study is the first test of seedling drought tolerance in *M. polymorpha*, a keystone species recruitment of which is critical for the conservation of Hawaii’s endemic biodiversity. To revisit our questions, we found that (1) *M. polymorpha* seedlings from a dry forest tolerate drought; (2) However, the degree of tolerance depends on the total amount of water, but varies little in response to the number of days between watering events for plants under press drought; (3) Multiple morphological and physiological traits were plastic in response to the treatments, and all four traits were correlated with seedling performance under either press or pulse drought, identifying them to be important mechanisms of drought tolerance.

Fig. 5 Influence of physiological traits on height (cm, **A–C**) and biomass (g, **D**) for *M. polymorpha* seedlings in different watering treatments. The control group (blue, inverted triangle) was watered daily to 100% field capacity (100FC); the 2-Day (orange, circle), 7-Day (red-orange, diamond), and 14-Day (brown, triangle) groups were watered to 80FC; the pulse drought group (green, square) received no water for 28 days. Corresponding linear models are represented with solid lines when statistically significant (Appendix 8): control = light blue, 2 day = orange, 7 day = red-orange, 14 day = brown, pulse drought = green. (Color figure online)



Seedling drought tolerance

We predicted that *M. polymorpha* seedlings would express drought tolerance given the broad distribution of this species and because we collected the seeds from a high elevation dry forest. Recent evidence of local adaptation to drought has been found in seedlings of this species across various Oahu populations (Barton et al. in review). As expected, the control group sustained the highest soil water content throughout the experiment, while the pulse drought group, which received no water for 28 days, experienced the lowest soil water content of all the groups in the experiment, but recovered when daily watering resumed. We also detected temporal variation in soil water content in all treatments, which may be due to changes in soil properties with infrequent watering and to increased water use by plants as they grew over the experimental

period. Using a similar experimental design, O'Brien et al. (2013) found that soil matric potential (a key determinant of water available to plants) was influenced by variations in the timing of watering, and O'Brien et al. (2015) reported that with infrequent watering, soil water is depleted more quickly as plants become larger.

Seedlings exposed to 80FC press drought at different frequencies of watering survived as well as and achieved similar final biomass as control plants, although they grew more slowly in terms of height. The 80FC groups varied little in their performance; plants in the 7-day group performed worse than the 2-day group, but this was due to initial size differences, which persisted throughout the experiment. In contrast to the 80FC groups, a 28-day pulse drought severely reduced seedling survival and growth. Similar findings have been reported by O'Brien et al.

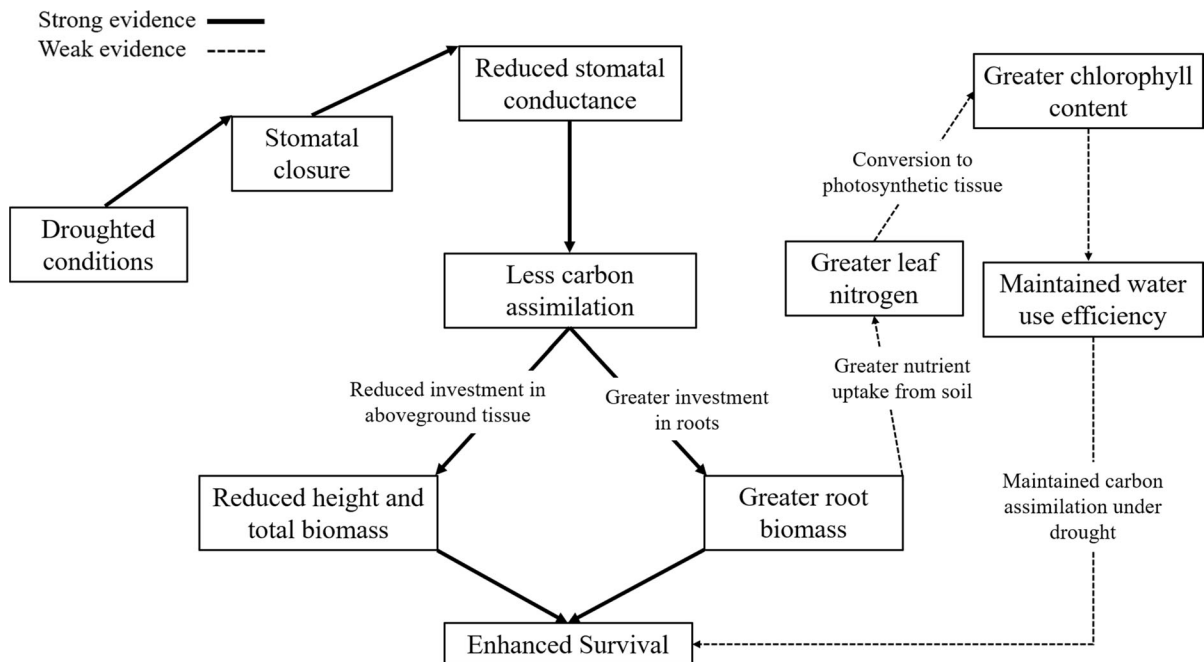


Fig. 6 Conceptual diagram of hypothesized adaptive response of *M. polymorpha* seedlings under drought stress. Thick solid lines represent significant statistical relationships among processes, while thin dashed lines represent relationships for which

we have weak statistical evidence. Fluorometry data are omitted for simplicity but see “Discussion” for importance of non-photochemical quenching under drought

(2013) who found that increasing the time between watering negatively influenced various seedling performance metrics. Increasingly longer droughts may thus prove catastrophic for *M. polymorpha* seedlings as they are unable to sustain growth and have lower survival. Our results highlight the importance for drought studies that simultaneously manipulate both the timing and intensity of drought. Although it can be challenging to design and execute experiments that mimic climate change scenarios (Beier et al. 2012), they are essential for developing robust predictions for future plant distributions.

Traits underlying seedling drought tolerance

Our secondary goal was to identify morphological and physiological traits associated with drought tolerance in *M. polymorpha* seedlings. Valladares and Sánchez-Gómez (2006) detected significant associations between physiological and morphological traits on seedling survival for 11 tree species grown under drought conditions, suggesting that seedling traits can indeed influence whole plant performance and be used

as a diagnostic tool to predict drought tolerance. *M. polymorpha* is slow-growing (Ostertag et al. 2009), and previous studies on mature trees have reported them to be conservative in water use (Kagawa et al. 2009; Cavaleri et al. 2014), expressing a wide range of pubescence and gas exchange rates among sympatric populations (Cordell et al. 1998; Hoof et al. 2008; DeBoer and Stacy 2013). Here, we report for the first time how drought influences *M. polymorpha* seedling physiology and morphology, which are summarized in a model presented in Fig. 6.

The model shows how drought conditions influence plant–water relations, potential carbon gain, and resource allocation (strength of relationships among traits are omitted but see Appendix 7 for correlations among variables and Fig. 5) in *M. polymorpha* seedlings. For surviving individuals, we observed low stomatal conductance compared to the control group, which likely limited carbon assimilation and triggered a shift in allocation toward enhanced belowground growth. We also observed weak evidence that the increase in root growth facilitated greater uptake of nitrogen, which was subsequently converted to chlorophyll. While greater nitrogen could

promote increased photosynthetic output and water-use efficiency we found no statistical evidence for this, suggesting that *M. polymorpha* may be increasing light use efficiency rather than carbon assimilation. Although photosynthetic capacity can increase under short and intense droughts (Brix 1962; Ruehr et al. 2009), it may not increase when droughts are more gradual and severe if seedlings can maintain enzymatic activity (Aaltonen et al. 2017). Kellomaki and Wang (1996) found that photosynthesis often declines due to stomatal limitation during early stages of drought, whereas under more chronic water stress, reductions in enzymatic activity may limit photosynthesis. Under drought, plants that increase allocation to leaf nitrogen may thus have higher light use efficiency (Susiluoto and Berninger 2007; Aaltonen et al. 2017). In our study, seedlings under press drought with high chlorophyll and high stomatal conductance grew fastest and also had high survival, whereas under the severe pulse drought, leaf carbon to nitrogen ratio was the only trait that significantly influenced growth performance. Thus, the maintenance of key physiological processes (e.g., stomatal conductance) under stressful periods of drought determines whole plant performance for *M. polymorpha* seedlings. Reductions in growth and survival under drought will surely limit the regenerative capacity of this widespread species under future drought conditions, and more work is needed to understand how the observed responses shift over the lifetime of *M. polymorpha* (i.e., a time series of physiological responses), and how performance under drought during early life stages influences drought tolerance later in life.

Instantaneous WUE varied by treatment but long-term water-use efficiency ($\delta^{13}\text{C}$) responded little to the treatments. $\delta^{13}\text{C}$ and instantaneous WUE may differ because instantaneous WUE depends on evaporative demand at the leaf-level and does not directly affect $\delta^{13}\text{C}$, which is integrated over the time of tissue synthesis (Seibt et al. 2008; Cernusak et al. 2013). $\delta^{13}\text{C}$ may also be influenced by plant metabolic processes, and the two indices may vary independently. For example, when assimilation rate is low due to low leaf metabolic rates, c_i should increase and $\delta^{13}\text{C}$ decrease (Farquhar et al. 1980). In contrast, when CO_2 assimilation rate is indirectly reduced by a reduction in stomatal conductance, c_i should decrease and $\delta^{13}\text{C}$ increase. The lack of an association between $\delta^{13}\text{C}$ and

instantaneous WUE may also have been partly affected by our inclusion of leaves that developed prior to separation into the treatments, which may have masked some of the treatment effects for the most water-stressed plants. Alternatively, it may be that during periods of drought stress, stomatal conductance was low and leaf production ceased entirely, resulting in an isotopic signature that reflected only favorable growth periods. We doubt the validity of this claim given the slow growth rates of *M. polymorpha* (Ostertag et al. 2009); however, leaf turnover rates would be needed to refute this hypothesis and determine whether seedlings underwent growth pulses.

Under periods of drought, plant growth may be diminished by the reduced uptake, transport, and redistribution of nitrogen and phosphorus (Cramer et al. 2009; Rouphael et al. 2012; Salazar-Tortosa et al. 2018) following stomatal closure. A potential adaptive response to drought is to invest more nitrogen in leaves to increase assimilation rates while reducing total leaf area and stomatal conductance (Farquhar et al. 2002). Our results partially supported this hypothesis; plants under pulse drought had lower stomatal conductance than well-watered plants, and nitrogen content slightly increased with drought severity. Nonetheless carbon:nitrogen varied in how it influenced growth performance among two of the watering treatments (Figs. 5, 6). These findings indicate that with changes in the frequency of watering events, *M. polymorpha* alter their leaf traits with lasting consequences for whole plant performance. Future studies should assess changes in the mineral nutrition of seedlings under drought, as calcium, chlorine, potassium, magnesium, nitrogen, sodium have been shown to accumulate in aboveground tissues for plants under drought stress (Mattson and Haack 1987).

The fluorometry data suggest that *M. polymorpha* seedlings under drought stress were also experiencing high light stress, and that the two simultaneous stressors influenced their performance. We observed a significant increase in maximum non-photochemical quenching (NPQ_{max}) for plants in the 14-day 80FC drought, and a significant decrease in light saturation (E_k) and maximum rate of electron transport (ETR_{max}) for plants in the 7-day 80FC drought. Changes in NPQ_{max} are indicative of the efficiency of heat dissipation from excess light energy and can result from damage to the photosystems or, as a mechanism

of photoprotection (Maxwell and Johnson 2000). Changes in fluorescence parameters have been found to correlate with drought stress (Conroy et al. 1986; Salazar-Tortosa et al. 2018; Wu et al. 2008), as stomatal closure reduces evaporative leaf cooling (Cook et al. 1964), which results in damage to photosynthetic machinery (Sharkey 2005). Despite their known associations and joint influence on potential carbon gain, simultaneous light and drought stress remain understudied (Valladares and Pearcy 2002; Niinemets and Valladares 2006). Salazar-Tortosa et al. (2018) found that seedlings of drought-sensitive pine species under drought conditions had lower WUE and lower photochemical efficiency, as well as lower growth and survival compared to more drought-tolerant species. In this study, plants under drought stress had a greater overall capacity to dissipate excess light energy than did well-watered plants but grew less when they exhibited reduced NPQ_{max} , revealing a significant interaction between drought and light stress with regards to growth performance for this widespread tropical tree. The ability to cope with multiple stressors is of critical importance for vulnerable plant life stages, and our findings indicate that preventing damage to its photosynthetic machinery is particularly important for *M. polymorpha*.

Compared to the physiological traits, leaf morphological traits demonstrated little plasticity in response to the watering treatments, and similar findings have been reported for adult *M. polymorpha* (Cordell et al. 1998) and for tree seedlings and saplings of various species from more arid climates (Matías et al. 2012; Chiatante et al. 2015; Maseda and Fernández 2016). Matías et al. (2012) found few significant associations between biomass and morphological/structural traits (e.g., leaf mass ratio, SLA, root mass ratio) and drought resistance (based on survival) in saplings of eight woody species. Chiatante et al. (2015) found no difference between control and drought-exposed groups of oak seedlings, in stem dry mass and leaf dry mass. Although we observed no effect of drought on specific leaf area, or pairwise differences in leaf thickness, we observed greater root biomass, which is likely to enhance water uptake for species in dry habitats (Markesteijn and Poorter 2009). A lack of morphological plasticity relative to physiological plasticity may result from a greater relative cost to

alter morphological traits (Bradshaw 1965), as morphological traits can be adjusted less rapidly than physiological traits. Furthermore, plants in resource-rich habitats where competition is strong may be more morphologically plastic whereas resource-poor environments often promote greater physiological plasticity (Grime 1977, 1986). Our study supports these patterns as it is likely that under the imposed drought, *M. polymorpha* seedlings had a greater capacity to alter their physiology as opposed to their morphology.

Conclusion

Our study provides evidence that not only the amount but also the timing of water availability can strongly influence seedling performance as well as underlying morphology and physiology. Recruitment of *M. polymorpha*, a keystone species in Hawaii and member of a Pan-Pacific genus (Wilson 1996), may be limited by future drought events that are severe and long-lasting. To better understand how *M. polymorpha* will regenerate under projected climate change scenarios will require further studies that include additional populations of *M. polymorpha*, from wet to dry forests, and expand the traits examined to include wood density, non-structural carbohydrate reserves (O'Brien et al. 2014; Corlett 2016) and mobilization, and pubescence. Our study highlights the importance of manipulating the magnitude and temporal variation of drought at the highly vulnerable seedling stage, explicitly considering how trait plasticity will influence drought tolerance.

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Appendix 1

Dates of experiment for *M. polymorpha* seedlings reared under various watering treatments

Day of experiment	Date	Action taken
D0	8-Jun-2016	Seedlings transplanted to greenhouse
D14	22-Jun-2016	Seedlings separated into watering treatments
D42	20-July-2016	Pulse drought ends. Plants in pulse drought group were then watered daily until end of experiment.
D69	16-Aug-2016	All plants harvested

Appendix 2

Number of measurements for each of 18 response variables, estimated for *M. polymorpha* seedlings in different watering treatments. See Appendix 3 for units

and descriptions of physiological traits. Photosynthesis, WUE, and Transpiration were not estimated for plants in the PD due to high mortality and small leaf size.

Type of response	Response variable	Watering treatment					Sampled multiple times
		Control	2 day	7 day	14 day	Pulse drought (PD)	
Performance metrics	Survival	19	20	19	20	12	No
	Total biomass	19	20	19	19	12	No
	Height	72	77	61	68	22	Yes
Physiological traits	Photosynthesis	3	3	3	5	0	No
	WUE	3	3	3	5	0	No
	Transpiration	3	3	3	5	0	No
	Stomatal conductance	94	106	89	92	81	Yes
	NPQ _{max}	16	17	15	17	4	No
	α	16	17	15	17	4	No
	ETR _{max}	16	17	15	17	4	No
	Ek	16	17	15	17	4	No
	Y ₀	16	17	15	17	4	No
	Chlorophyll content	15	14	14	16	1	No
	$\delta^{13}\text{C}$	10	10	10	10	10	No
	%N	10	10	10	10	10	No
Morphological traits	Root biomass, and root:shoot biomass	18	20	19	19	12	No
	Leaf thickness	16	17	15	17	4	No
	SLA	19	20	19	19	12	No

Appendix 3

Response variables and corresponding units measured on *M. polymorpha* seedlings in different watering treatments.

Type of response	Response variable (units)	Information provided
Performance Metrics	Survival (number of plants)	Whole plant performance
	Total (dry weight) biomass (g)	
	Height (cm)	
Physiological Traits	Net photosynthesis (A , $\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$)	Assimilation rate; potential carbon gain
	Instantaneous water-use efficiency (WUE, AE^{-1})	Potential carbon gain relative to water loss
	Transpiration rate (E , $\text{mmol H}_2\text{O m}^{-2} \text{ s}^{-1}$)	Leaf water loss
	Stomatal conductance (g_s , $\text{mol H}_2\text{O m}^{-2} \text{ s}^{-1}$)	Leaf water loss and potential carbon gain
	Maximum non-photochemical quenching (NPQ_{max} , relative units)	$(F_m^0 - F_m')/F_m'$; protective mechanism where excess absorbed light energy is dissipated as heat
	Quantum efficiency (α , mol electrons/mol photons)	Ratio of absorbed CO_2 per photons of incident light; maximal light energy utilization the photosynthetic apparatus of a plant achieves
	Maximum electron transport capacity (ETR_{max} , $\mu\text{mol electrons m}^{-2} \text{ s}^{-1}$)	Rate at which electrons move through the photosynthetic chain; photochemical response
	Minimum saturating irradiance (E_k , $\mu\text{mol photons m}^{-2} \text{ s}^{-1}$)	$\text{ETR}_{\text{max}}\alpha$; indicator of photosynthetic efficiency and potential for damage or light utilization potential

Type of response	Response variable (units)	Information provided
	Quantum yield (Y_0 , $\text{mol CO}_2 \text{ mol photons}^{-1}$)	$Y_0 = (F_m - F_0)/F_m$; maximal photochemical PSII yield
	Chlorophyll content (relative values)	Indicator of photosynthetic capacity and carbon gain
	$\delta^{13}\text{C}$ (‰)	Integrated water-use efficiency; potential carbon gain relative to water loss
	%N and C:N ratio (unitless)	Leaf nitrogen content (%) on a mass basis, and carbon to nitrogen ratio. Indicators of relative investment in photosynthetic tissue.
Morphology	Root:shoot biomass (unitless) and root biomass (g)	Regulation of soil water uptake
	Leaf thickness (mm)	Relative investment in photosynthetic tissue and resistance to desiccation
	Specific leaf area (SLA , $\text{cm}^2 \text{ g}^{-1}$)	Relative investment in photosynthetic tissue and resistance to desiccation

References: Lawlor (1987), Maxwell and Johnson (2000), and Behrenfeld et al. (2004)

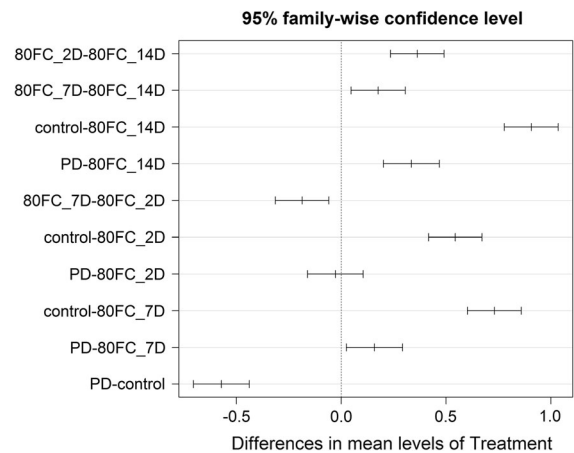
Appendix 4

Metrosideros polymorpha seedlings. Images on left captured June 28, 2017 by A. Westerband depict K. Bogner measuring chlorophyll content with the SPAD 502 Plus Chlorophyll Meter. Images on right captured on January 30, 2018 by Andrea Westerband, and thus do not represent individuals used in the present study. Pictures on right were thus included for a more accurate depiction of scale.



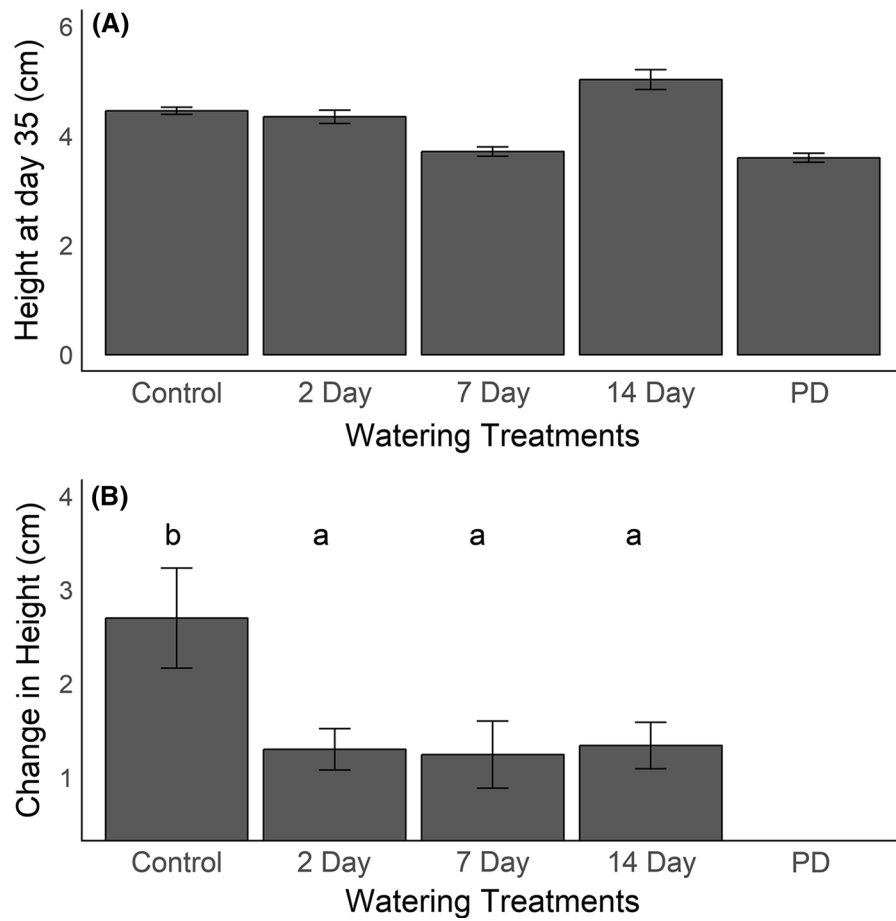
Appendix 5

Two-way comparisons of mean soil water content readings for *M. polymorpha* seedlings in different watering treatment groups. 95% family-wise confidence level. Statistically significant comparisons do not overlap with the vertical dotted line. Line segments to the left of the vertical dotted line indicate that the difference between the mean soil water of the two groups is negative, and those to the right of the line have a positive difference in their mean. All comparisons were statistically significant ($P < 0.05$) except for the 2-day (2D) watering by pulse drought (PD) comparison.



Appendix 6

Seedling height measured on day 35 and change in height for *M. polymorpha* across watering treatments ($N = 37$). 'PD' is 28-day pulse drought. Means $\pm$ SE are shown for all traits.



Appendix 7

Correlation matrices for physiological traits and morphological traits measured in *M. polymorpha*

seedlings. Boldface values indicate statistically significant correlations at $\alpha = 0.1$. A Holm correction was used to adjust *P* values for multiple comparisons.

	Photosynthesis	WUE	Transpiration	Stomatal conductance	ETR _{max}	E _k	Y ₀	α	Chlorophyll	NPQ _{max}	%N	C:N	δ ¹³ C
WUE	0.44	1.00											
Transpiration	0.51	– 0.50	1.00										
Stomatal conductance	0.12	0.20	0.03	1.00									
ETR _{max}	0.30	– 0.06	0.33	0.42	1.00								
E _k	0.36	– 0.07	0.39	0.42	0.90	1.00							
Y ₀	– 0.03	0.05	– 0.08	– 0.06	0.08	– 0.08	1.00						
α	– 0.03	0.06	– 0.06	0.20	0.54	0.15	0.26	1.00					
Chlorophyll	0.68	0.48	0.16	0.35	0.37	0.35	0.13	0.06	1.00				
NPQ _{max}	0.18	0.34	– 0.14	– 0.21	– 0.41	– 0.44	– 0.19	0.01	– 0.24	1.00			
%N	– 0.01	0.54	– 0.47	0.17	0.23	0.22	– 0.23	0.10	0.29	– 0.12	1.00		
C:N	0.18	– 0.42	0.46	– 0.20	– 0.21	– 0.19	0.22	– 0.10	– 0.29	0.11	– 0.99	1.00	
δ ¹³ C	– 0.96	– 0.46	– 0.17	– 0.41	– 0.12	– 0.27	0.22	0.22	– 0.29	0.33	0.26	– 0.24	1.00
SLA													
Leaf thickness													
SLA					– 0.22								
Root:shoot					– 0.07								0.25
Root biomass					0.02								0.77

Appendix 8

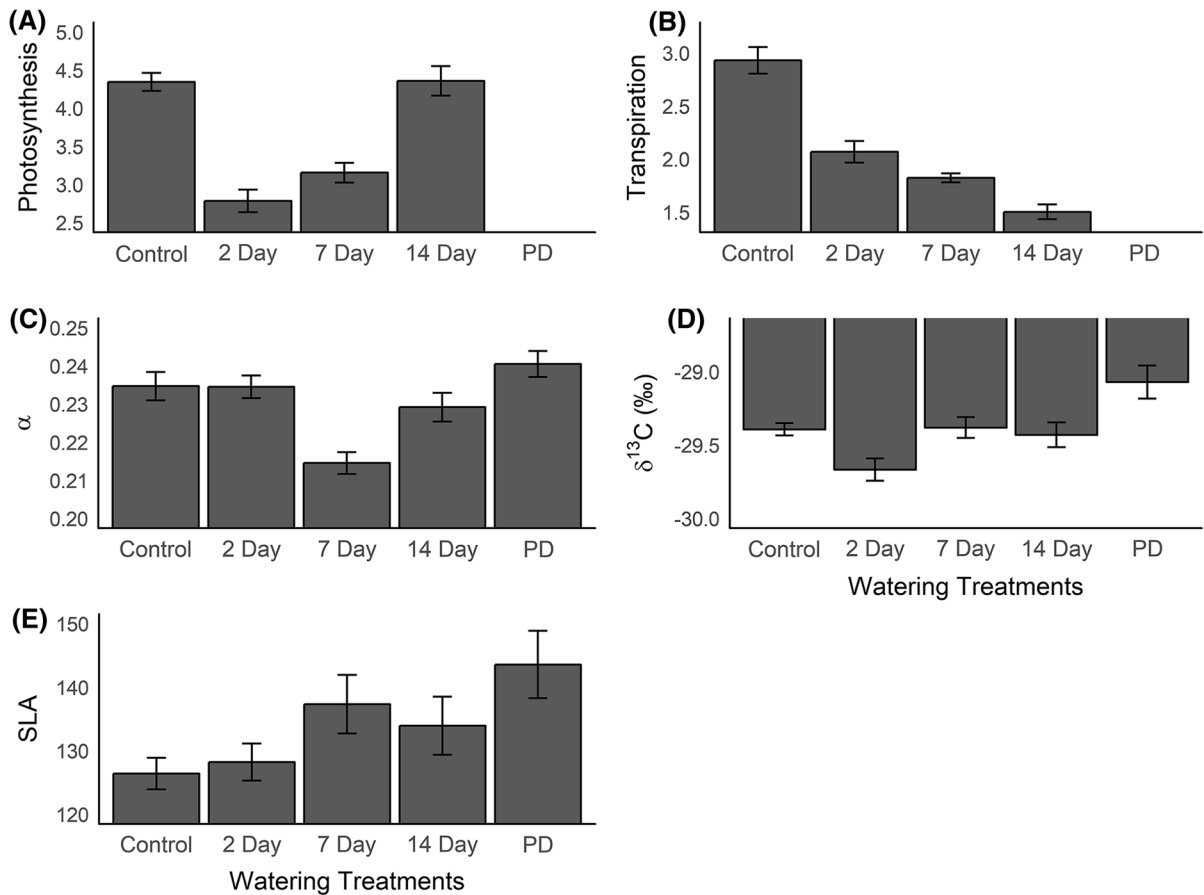
Slope coefficient and R^2 (in parenthesis) for linear regression models testing seedling growth performance as a function of leaf traits. Statistically

significant models ($P < 0.05$) are indicated with boldface, and only significant models are shown in Fig. 5. MS = marginally significant, $\alpha < 0.1$. NA indicates insufficient data to test model.

Linear regression model	Treatment group				
	Control	2 day	7 day	14 day	Pulse drought
Height ~ stomatal conductance	0.00005 (0.001)	0.0003 (0.04)^{MS}	0.0007 (0.16)	0.0009 (0.18)	0.0002 (0.04)
Height ~ chlorophyll content	– 0.01 (0.03)	– 0.003 (0.005)	0.05 (0.39)	0.03 (0.22)^{MS}	NA
Height ~ NPQ _{max}	0.27 (0.10)	– 0.21 (0.13)	– 0.07 (0.01)	– 0.59 (0.42)	NA
Total biomass ~ Leaf carbon:nitrogen	0.08 (0.26)	0.03 (0.03)	0.21 (0.82)	– 0.01 (0.002)	0.16 (0.46)

Appendix 9

Functional traits of *M. polymorpha* seedlings that were not influenced by the watering treatments. Means $\pm$ SE are shown for all traits. Units: (A) Photosynthesis, $\mu\text{mol m}^{-2} \text{s}^{-1}$; (B) transpiration, $\text{mmol H}_2\text{O m}^{-2} \text{s}^{-1}$; (C) α , mol/mol; (D) $\delta^{13}\text{C}$, ‰; (E) SLA; $\text{cm}^2 \text{g}^{-1}$



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